

Temperature-Based DUAL SAFETY SYSTEM For Smart Homes

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A temperature-based dual safety system addresses common household risks through intelligent control. Built on an Arduino Uno, it operates in two selectable modes via a slide switch.

In one mode, the system closes the gas cylinder knob when milk temperature exceeds a preset threshold, preventing overflow. In the other, it switches off an electric heater rod when water temperature rises beyond a limit, preventing burns and electric shock. The threshold is set using a 4×4 keypad, while an LCD displays real-time temperature along with the set value.

Designed for everyday use,

the device suits households with elderly users and children. It is cost-effective, easy to install, and uses components listed in the Bill of Materials table on next page.

Circuit and working

Fig. 1 shows the temperature-based dual safety system for smart homes. The system is built around an Arduino Uno (MOD1), a DS18B20 digital temperature sensor (S1), a 4×4 keypad (MOD2), a 16×2 LCD (LCD1), a servo motor (M1), a 5V relay (RL1), and supporting components.

The Arduino Uno acts as the main controller, processing sensor data and managing control

operations. A relay controls the heater, while a servo motor operates the gas knob. Alternatively, an electronic gas valve or LPG solenoid valve may be used with minor code modifications.

A slide switch enables mode selection. For alerts, an LED and a buzzer provide visual and audio indication when the temperature exceeds the preset threshold. The relay is driven by a BC547 transistor with a flyback diode for protection, enabling safe control of an external AC load such as a heater.

During operation, the Arduino monitors the temperature, compares it with the set value, and triggers outputs such as the buzzer, LED,

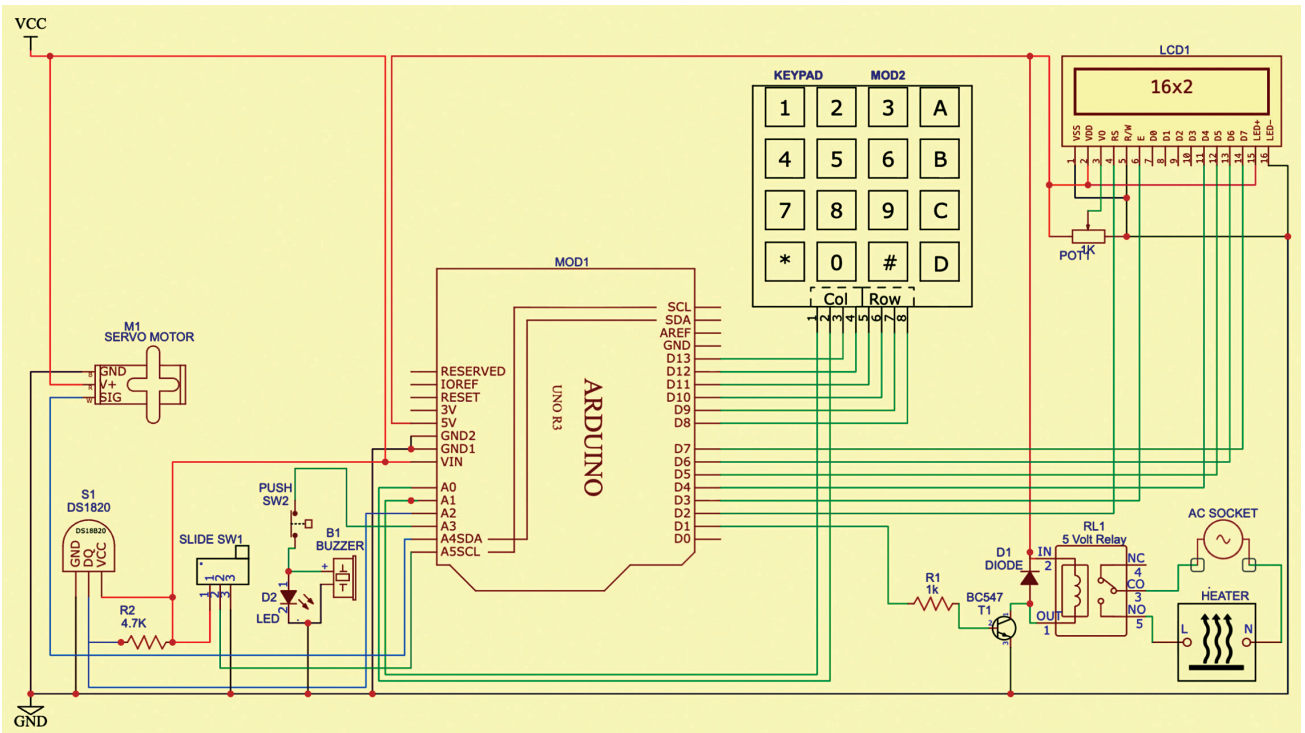


Fig. 1: Circuit diagram